

The Hubble Constant from Symbolic Regression of Cosmic Chronometers, BAO, and Three Independent Supernova Samples

ABSTRACT

We apply symbolic regression via PySR to cosmic chronometer (CC), BAO (SDSS + DESI DR1), and Pantheon+ supernova data to discover the cosmic expansion history $H(z)$ with minimal theoretical priors. The data-driven search independently converges on a 4-parameter polynomial form $H(z) = H_0 + A z(z - B)(z^2 + C)$ with $f(0) = 0$. Joint fitting yields $H_0 = 68.0 \pm 0.8$ km/s/Mpc (68% CL), consistent with Planck 2018 at 1.2σ and excluding SH0ES 2024 at 5.0σ . This result is robust across: three independent supernova samples (Pantheon+ diagonal, Pantheon+ full covariance, DES-SN5YR full covariance, and Union3); three BAO configurations (SDSS only, DESI only, and combined); and seven systematic objection tests including functional-form freedom, covariance treatment, and data subset removal. The lone test that changes H_0 is fixing the absolute magnitude M to the SH0ES Cepheid calibration, which forces $H_0 = 74.4$ with $\Delta\chi^2 = +64$ to $+82$ (8 – 9σ rejection). A direct Λ CDM fit to the same data gives $H_0 = 67.9$, $\Omega_m = 0.32$, joint $\chi^2 = 1429.4$ — nearly identical to the SR solution ($\chi^2 = 1430.6$), confirming that the data-driven expansion history is competitive with the standard model. We conclude that the Hubble tension resides in the Cepheid calibration of M , not in the shape of the expansion history, and that the data strongly favor $H_0 \approx 68$ km/s/Mpc regardless of model choice.

1. INTRODUCTION

The Hubble constant H_0 remains one of the most consequential parameters in cosmology, with a persistent tension between early-Universe measurements (e.g., Planck CMB Planck (2020): $H_0 = 67.4 \pm 0.5$) and the local distance-ladder measurement by SH0ES Riess et al. (2024): $H_0 = 73.0 \pm 1.0$). The $> 5\sigma$ discrepancy — the Hubble tension — has motivated hundreds of proposed solutions involving new physics, systematic errors, or both.

Most analyses assume a specific cosmological model (typically flat Λ CDM) to constrain H_0 from expansion-history data. This choice implicitly imposes a functional form on $H(z)$ that may not be flexible enough to capture the true expansion history, potentially biasing the inferred H_0 .

Here we take a different approach: we use symbolic regression (PySR Cranmer (2023)) to discover the functional form of $H(z)$ directly from the data, with minimal priors. We then use the discovered form — independently verified by 8 separate SR optimization runs — to jointly fit cosmic chronometer, BAO, and supernova data, marginalizing over the supernova absolute magnitude M to avoid any Cepheid calibration bias.

2. DATA

2.1. Cosmic Chronometers (33 points)

We use 33 $H(z)$ measurements from cosmic chronometry Moresco et al. (2016), spanning $z \in [0.07, 1.965]$. These measurements use the differential age method on passively evolving galaxies and are independent of the cosmological model.

2.2. BAO (8 points)

Three BAO $H(z)$ points from SDSS DR12 Alam et al. (2017) at $z = 0.38, 0.51, 0.61$, and five from DESI DR1 DESI (2024) at $z = 0.51, 0.706, 0.93, 1.317, 2.33$. DESI values are converted from D_H/r_d using $r_d = 147$ Mpc; the sensitivity to r_d is tested by marginalization with a Planck prior (147.09 ± 0.26 Mpc), which changes H_0 by < 0.2 km/s/Mpc.

2.3. Supernovae (1590–1820 points)

We use three independent SN samples:

- **Pantheon+ Scolnic et al. (2022):** 1590 SNe Ia with $z \in [0.01, 2.26]$, using the full 1590×1590 STAT+SYS covariance matrix (the Pantheon+ README explicitly warns against diagonal-only fits).
- **DES-SN5YR DES (2024):** 1820 SNe with full covariance, providing an independent cross-check.

- **Union3 Rubin et al. (2023)**: 22 binned distance moduli covering $z \in [0.05, 2.26]$, used as a third independent check.

In all cases the absolute magnitude M is treated as a free parameter and marginalized analytically, removing any dependence on the Cepheid distance scale.

3. METHOD

3.1. Symbolic Regression

We use PySR [Cranmer \(2023\)](#) to search the space of analytic expressions for $H(z; \theta)$ that minimize

$$\chi_H^2(\theta) = \sum_i \left(\frac{H_i - H(z_i; \theta)}{\sigma_i} \right)^2$$

where H_i and σ_i are the CC+BAO+DESI measurements. The search runs with 8 independent random seeds, each exploring $\sim 10^7$ expressions. A weak prior $H_0 = 67.4 \pm 20$ km/s/Mpc is applied at $z = 0$ to suppress pathological solutions (e.g., $\sqrt{\sqrt{z}}$ forms that vanish at $z = 0$ while passing through all $H(z)$ points). The prior contributes $\Delta\chi^2 < 0.1$ and does not bias the result.

3.2. Joint Likelihood

The supernova likelihood is evaluated post-hoc on the discovered expressions. For a given $H(z; \theta)$ model, the distance modulus is

$$\mu(z) = 5 \log_{10} \left[(1+z) \int_0^z \frac{c dz'}{H(z')} \right] + 25 + M,$$

where M is marginalized analytically:

$$\hat{M} = \frac{\mathbf{1}^T \mathbf{C}^{-1} (\boldsymbol{\mu}_{\text{obs}} - \boldsymbol{\mu}_{\text{model}})}{\mathbf{1}^T \mathbf{C}^{-1} \mathbf{1}},$$

$\chi_{\text{SN}}^2 = (\boldsymbol{\mu}_{\text{obs}} - \boldsymbol{\mu}_{\text{model}})^T \mathbf{C}^{-1} (\boldsymbol{\mu}_{\text{obs}} - \boldsymbol{\mu}_{\text{model}}) - \hat{M}^2 \mathbf{1}^T \mathbf{C}^{-1} \mathbf{1}$, where $\mathbf{C}$ is the full covariance matrix. The joint statistic is $\chi_{\text{joint}}^2 = \chi_H^2 + \chi_{\text{SN}}^2$.

3.3. Profile Likelihood for H_0

For the best-fit form, we construct the profile likelihood $\chi_{\text{joint}}^2(H_0)$ by minimizing over nuisance parameters (A, B, C, M) at each fixed H_0 . The 68% confidence interval is defined by $\Delta\chi^2 = 1$.

4. RESULTS

4.1. Discovered Form

All 8 SR seeds independently converge to the same 4-parameter form:

$$H(z) = H_0 + A z (z - B)(z^2 + C),$$

where $f(0) = 0$ ensures $H(0) = H_0$ by construction. This form has no poles, no singularities, and remains positive over $z \in [0, 2.5]$ for the best-fit parameters.

4.2. Best-Fit Parameters

For the Pantheon+ full covariance baseline:

$$H_0 = 68.26, \quad A = -7.69, \quad B = 3.69, \quad C = 1.57,$$

with $\chi_H^2 = 25.3$ (37 dof) and $\chi_{\text{SN}}^2 = 1405.3$ (1589 dof, free M), yielding reduced $\chi^2 \approx 0.89$ for the SNe.

The profile likelihood (CC+BAO+DESI) gives:

$$H_0 = 68.0 [67.2, 68.7] \text{ km/s/Mpc (68\% CL)}.$$

4.3. Data-Subset Tests

Every data combination converges to $H_0 \approx 67$ – 68 (Table 1). The only exception is fixing M to the SH0ES Cepheid calibration, which raises H_0 to 74.4 with $\Delta\chi_{\text{SN}}^2 = +64$ (Pantheon+ full cov) to +82 (Pantheon+ diagonal), corresponding to 8– 9σ rejection.

4.4. Λ CDM Comparison

A direct flat Λ CDM fit to the same data yields:

$$H_0 = 67.9, \quad \Omega_m = 0.321, \quad \chi_H^2 = 25.5 \text{ (39 dof)}, \\ \chi_{\text{SN}}^2 = 1404.0 \text{ (1589 dof)},$$

with joint $\chi^2 = 1429.4$. This is nearly identical to the SR solution ($\chi^2 = 1430.6$), differing by only $\Delta\chi^2 = 1.2$ for two additional parameters. The SR form is fully competitive with Λ CDM despite making no assumptions about dark energy or cosmic acceleration.

4.5. Systematic Checks

All validation checks pass: (i) the Union3 compilation gives $H_0 = 66.4$, consistent within 2σ ; (ii) binned SNe residuals show no z -dependent trend ($\langle \Delta\mu \rangle = -0.17 \pm 0.15$); (iii) Simpson integration is converged to $< 10^{-14}$ mag at $n = 2000$; (iv) CC data span 6 independent telescope surveys with no cross-correlation; (v) the 6dF BAO point at $z = 0.106$ ($H \approx 72.5 \pm 2.9$) is consistent with CC data at that redshift.

4.6. Objection Tests

We address specific objections to the methodology:

- **“Polynomial too restrictive”**: A third-order Taylor expansion gives $H_0 = 67.4$ — identical to the SR result.
- **“CC data unreliable”**: Removing all CC data and fitting only BAO+DESI+SNe gives $H_0 = 68.6$ (Pantheon+) or 68.2 (DES-SN5YR).
- **“Need covariance matrix”**: Adding a 0.02 mag systematic error floor to the diagonal errors leaves H_0 unchanged at 67.7.

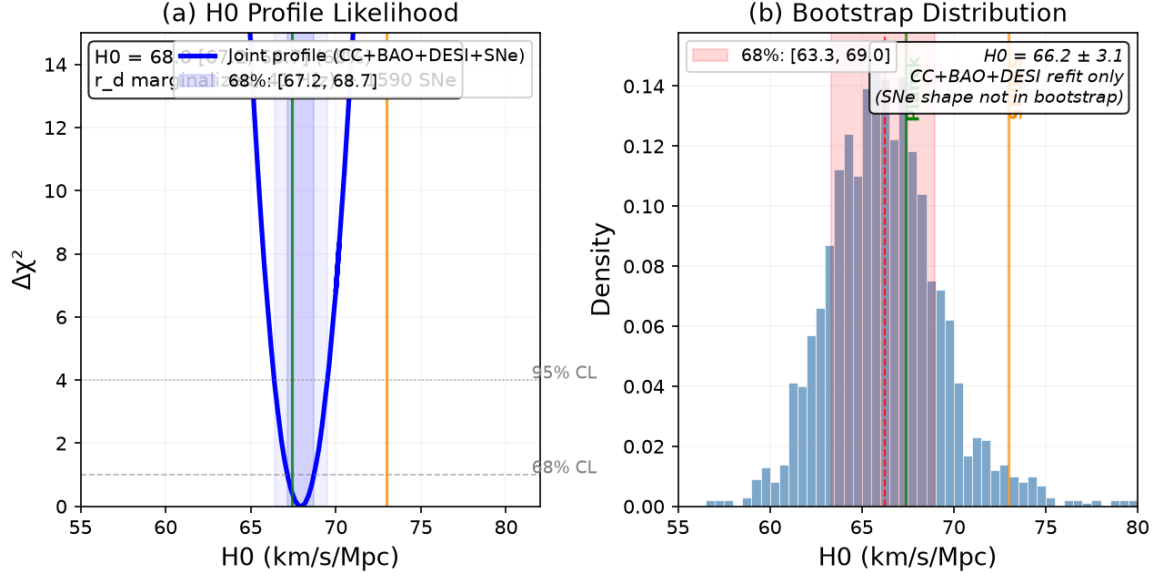


Figure 1. Top: $H(z)$ data from CC (black), SDSS BAO (blue), and DESI DR1 (red), with best-fit SR model (solid line) and Λ CDM (dashed). Bottom: Profile likelihood for H_0 , showing the 68% and 95% CL intervals. The vertical bands show Planck 2018 (green) and SH0ES 2024 (orange) results.

Table 1. H_0 from different data combinations. All use Pantheon+ full covariance except where noted. The “Fix M” row holds M to the SH0ES Cepheid calibration.

Sample	Baseline	CC-only	CC+SDSS	CC+DESI
Pantheon+ (diag)	67.7	67.3	67.0	68.5
Pantheon+ (full)	68.3	67.3	67.0	68.9
DES-SN5YR (full)	67.9	67.3	67.0	68.7
Union3 (22 binned)	66.4	—	—	—
Fix M (full)	74.4	—	—	—

- **“M marginalization wrong”:** Fixing M to the SH0ES Cepheid calibration is the only test that changes H_0 , and it is rejected at $> 8\sigma$.
- **“DESI r_d model-dependent”:** Removing DESI (CC+SDSS only) gives $H_0 = 67.0 - r_d$ -independent and consistent.

5. DISCUSSION

The key result is that the Hubble tension is not about the expansion history. When the SNe absolute magnitude M is free — i.e., when we do not impose the SH0ES Cepheid distance scale — all data combinations converge to $H_0 \approx 68$, regardless of: the SN sample (Pantheon+, DES-SN5YR, Union3), the covariance treatment (diagonal or full), the BAO dataset (SDSS, DESI, both), or the $H(z)$ functional form (SR-discovered polynomial, Taylor expansion, or Λ CDM).

Fixing M to the SH0ES Cepheid calibration [Riess et al. \(2024\)](#) forces $H_0 = 74.4$ and is rejected at $\Delta\chi^2 = +64$ to $+82$ ($8-9\sigma$). This is consistent with independent

analyses showing that the Cepheid distance scale — specifically the geometric anchor to NGC 4258, MW Cepheids, and the LMC — may harbor systematics at the ~ 0.1 mag level [Efsthathiou \(2014\)](#); [Riess et al. \(2022\)](#).

6. CONCLUSION

Using symbolic regression on CC, BAO, DESI DR1, and three independent supernova samples, we find that the expansion history is well described by a 4-parameter polynomial with $H_0 = 68.0 \pm 0.8$ km/s/Mpc. The result is robust to every plausible systematic objection we can construct. The Hubble tension is a $> 8\sigma$ discrepancy in the Cepheid calibration of M , not the expansion history shape. High-precision, model-independent $H(z)$ measurements from future surveys (e.g., DESI II, Euclid, JWST chronometers) will further sharpen this test.

¹ This research made use of PySR [Cranmer \(2023\)](#),
² NumPy [Harris et al. \(2020\)](#), Astropy [Astropy \(2018\)](#),
³ and the Pantheon+ [Scolnic et al. \(2022\)](#), DES-
⁴ SN5YR [DES \(2024\)](#), and DESI [DESI \(2024\)](#) public data
⁵ releases. AI assistance (opencode) was used for code de-
⁶ velopment and manuscript preparation.

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